MYSQL FOUNDATIONS · BEFORE YOU BEGIN

Set Up MySQL for the Course

About half an hour to do **before the first session**: install MySQL 9.7 LTS, give its administrator account a strong password, make a course folder and run a short check that tells you (and your instructor) whether your laptop is ready. These are the same steps as Lesson 1.2 on the course site (rays-mysql.netlify.app, lesson "Installing MySQL"), which shows every step with its expected output. Session 1 then spends its time on databases instead of progress bars.

⏱️ 20–40 min, mostly downloading | No SQL needed yet | Windows · Mac · Linux · Docker | Free software

> **PLEASE DO THIS AT HOME, NOT IN CLASS**
>
> The installer is a download of a few hundred megabytes, and venue Wi-Fi is rarely up to twenty people doing that at once. If a step fails, keep going with the others and bring the **exact error message** (a screenshot is fine) to Session 1: its first half hour includes an environment check, and there is an install clinic after the break.

WHAT YOU NEED

- **A laptop** (Windows 10/11, macOS, or Linux) and its **charger**, with **permission to install software** (administrator rights) and about **1.5 GB of free disk space**. Chromebooks, tablets and locked work laptops cannot run the course comfortably: talk to your instructor first.
- **A terminal**: PowerShell on Windows, Terminal on macOS, your usual terminal on Linux. You will type a few commands into it; the course explains each one.
- **A plain-text editor** for `.sql` files. **VS Code** (free) works well. Not Word or TextEdit in rich-text mode: their curly quotes break SQL.
- **The environment check**: three small files from `labs/module-01/l1_0-orientation/` that your instructor sends with this guide (`check_environment.sql`, `check_environment.sh`, `check_environment.ps1`).
- **A learning journal**: a plain text file or a paper notebook. Every lesson ends with a journal prompt.
- **A password manager** (or a safe place) for the MySQL root password you are about to choose.

AM I READY FOR THIS COURSE?

MySQL Foundations starts from zero: no SQL and no programming are needed. It builds on everyday spreadsheet and computer skills. You are ready if you can:

- ☐ Filter a spreadsheet to the rows where a column has a certain value, and sort it by a column
- ☐ Say what "city is Chicago AND total over 20" keeps that "city is Chicago OR total over 20" does not
- ☐ Write a formula column such as price times quantity, and notice what happens with an empty cell
- ☐ Install a program, find a downloaded file, and say which folder a file is in (for example `Downloads/bookstore.sql`)
- ☐ Open a terminal, and copy a command into it exactly

Two or more unchecked? Come anyway and tell your instructor. Session 1 includes a short, ungraded diagnostic on these same skills, so your instructor knows where to help.

SET UP IN 5 STEPS

1 Choose the version and one path

~2 min

The course uses **MySQL 9.7 LTS** (long-term support), the free **MySQL Community Server**. Every output on the lesson pages was produced on 9.7, so yours will match.

- **Ask for 9.7 by name.** After 9.7, MySQL switched to calendar version numbers (year.month): its frequent **innovation releases** are numbered like **26.x**, and "latest" on a download page, `brew install mysql` or `mysql:latest` usually means one of those. They work, but some outputs will differ from the pages.
- **MySQL 8.4 LTS** also runs the course almost everywhere. **MySQL 8.0**, which many old tutorials show, is past its end of life: if a package offers you 8.0, use the steps below instead.
- **XAMPP ships MariaDB**, not MySQL, even though its button says "MySQL". **MAMP ships MySQL**, usually an older series. Both can follow most of the course (the lessons' version notes explain the differences), but for matching outputs install MySQL 9.7 itself.
- **One server per port.** If MySQL, MariaDB, XAMPP or MAMP is already installed and running, stop it first: two servers cannot both listen on port 3306.

YOUR COMPUTER	RECOMMENDED	ALTERNATIVE
Windows 10/11	MSI installer + MySQL Configurator	Docker Desktop, or the Linux steps inside WSL
macOS	Homebrew `mysql@9.7` or the DMG package	Docker Desktop
Ubuntu / Debian	MySQL's APT repository	Docker
Fedora / RHEL / Rocky / Alma	MySQL's Yum repository	Docker or Podman
Already use Docker	the `mysql:9.7` image	

Do **one** of the paths in Step 2, then continue with Step 3.

2 Install MySQL 9.7

~15-30 min

WINDOWS: THE MSI INSTALLER AND MYSQL CONFIGURATOR

1. **Download.** At **dev.mysql.com/downloads/mysql** (MySQL Community Server), choose **9.7.x LTS** and **Microsoft Windows**, then the **Windows (x86, 64-bit), MSI Installer**. "No thanks, just start my download" skips the account sign-up.
2. **Install.** Run the `.msi`, accept the license, choose **Typical**, confirm the administrator prompt. Leave "run MySQL Configurator" checked at the end. (If it asks for the Visual C++ Redistributable, install the current x64 one from Microsoft and run the MSI again.)
3. **Configure.** In MySQL Configurator: config type **Development**, port **3306**; type a **strong root password** twice and store it; keep "Configure MySQL Server as a Windows Service" and "Start at System Startup" checked and note the service name (something like `MySQL97`); skip the sample databases; **Execute**, wait for the green checks, **Finish**.
4. **Put the client on your PATH.** The MSI does not. Start ▸ "Edit the system environment variables" ▸ Environment Variables… ▸ your user **Path** ▸ Edit ▸ New, and add `C:\Program Files\MySQL\MySQL Server 9.7\bin`. OK three times, then open a **new** PowerShell window and check with `mysql --version`.

Start and stop later: the Services app, or `Get-Service MySQL*`, `Start-Service MySQL97`, `Stop-Service MySQL97` in an administrator PowerShell.

MACOS: HOMEBREW (OPTION A) OR THE DMG PACKAGE (OPTION B)

Option A, Homebrew. Ask for the 9.7 series by name (plain `brew install mysql` gives an innovation release). Versioned formulae are "keg-only", so add their folder to your PATH:

```
brew install mysql@9.7
echo 'export PATH="$(brew --prefix)/opt/mysql@9.7/bin:$PATH"' >> ~/.zshrc
source ~/.zshrc
mysql --version
brew services start mysql@9.7
```

Homebrew creates `root` **without a password**: Step 3 fixes that. Stop it later with `brew services stop mysql@9.7`.

Option B, the DMG. At dev.mysql.com/downloads/mysql choose **9.7.x LTS**, **macOS**, and the **DMG Archive** for your processor (**ARM, 64-bit** for Apple silicon, **x86, 64-bit** for Intel; Apple menu ▸ About This Mac tells you). Run the `.pkg`, set a strong root password near the end, then put the programs on your PATH:

```
echo 'export PATH="/usr/local/mysql/bin:$PATH"' >> ~/.zshrc
source ~/.zshrc
mysql --version
```

The DMG starts MySQL automatically and adds a MySQL pane to System Settings for starting and stopping it.

LINUX: MYSQL'S APT OR YUM REPOSITORY

Your distribution's own package is usually **not** MySQL 9.7. Check with `apt policy mysql-server`:

DISTRIBUTION PACKAGE	WHAT YOU GET
`mysql-server` on Ubuntu 22.04 and 24.04	MySQL 8.0 (past its end of life)
`mysql-server` on Ubuntu 26.04	MySQL 8.4 LTS
`default-mysql-server` on Debian	MariaDB, not MySQL
MySQL's own APT or Yum repository	the series you pick: **9.7 LTS**

Ubuntu and Debian. Download `mysql-apt-config_…_all.deb` from **dev.mysql.com/downloads/repo/apt**. Installing it opens a text menu: under the server entry choose **9.7 LTS**, then Ok. Set a strong root password when the install asks.

```
sudo dpkg -i ~/Downloads/mysql-apt-config_*_all.deb
sudo apt update
sudo apt install mysql-server
systemctl status mysql        # look for "active (running)"
```

Fedora, RHEL, Rocky, AlmaLinux. Download the repository package for your release from **dev.mysql.com/downloads/repo/yum**, install it with `sudo dnf install ./mysql*-community-release-*.rpm`, make sure the 9.7 LTS series is the enabled one, then:

```
sudo dnf install mysql-community-server
sudo systemctl enable --now mysqld
sudo grep 'temporary password' /var/log/mysqld.log
```

The first start writes a random **temporary** root password to that log: you change it in Step 3. In **WSL**, follow the Ubuntu steps inside your Linux terminal (on an old WSL without systemd: `sudo service mysql start`).

DOCKER: ANY OPERATING SYSTEM

With Docker Desktop (Windows, macOS) or Docker Engine (Linux), the official image is the cleanest install. Choose your own password (on Windows PowerShell, put the command on one line):

```
docker run --name mysql97 \
  -e MYSQL_ROOT_PASSWORD='choose-a-strong-password' \
  -p 3306:3306 \
  -v mysql97-data:/var/lib/mysql \
  -d mysql:9.7
docker logs mysql97          # wait for "ready for connections ... port: 3306"
docker exec -it mysql97 mysql -u root -p
```

- Use the tag `mysql:9.7`, not `mysql:latest` (an innovation release). Port 3306 taken? `-p 3307:3306`.
- The password counts only on the **very first start** of the `mysql97-data` volume; the data survives `docker stop`, `docker start mysql97` and even `docker rm`.
- The image contains the `mysql` client (the last line above). A client installed on the laptop connects with `mysql -h 127.0.0.1 -P 3306 -u root -p`: use `127.0.0.1`, not `localhost`.
- After a reboot, open Docker Desktop and run `docker start mysql97`.

3 Secure the root account, then log in

~5 min

`root` is MySQL's administrator. It must have a strong password; your instructor will never ask you to share it.

YOU INSTALLED WITH	ROOT PASSWORD NOW	WHAT TO DO
Windows MSI, macOS DMG, Docker	The one you set	Nothing more
MySQL APT repository	The one you set	Optionally `sudo mysql_secure_installation`
Homebrew	None	Run `mysql_secure_installation` (answer y to removing anonymous users, remote root and the test database)
Yum repository	Temporary, from the log	Log in with it and change it (below)
Ubuntu's own package	None: only `sudo mysql` works	Set a password (below), or better, use MySQL's APT repository

To set the password by hand, open the client (`sudo mysql`, or `mysql -u root -p` with the temporary password) and run, with your own password inside the quotes:

```
ALTER USER 'root'@'localhost'
    IDENTIFIED WITH caching_sha2_password BY 'your-strong-password-here';
```

Then log in and ask the server who it is. Nothing appears while you type the password; that is normal. Leave with `exit`.

```
mysql -u root -p
SELECT VERSION();
```

```
+-----------+
| VERSION() |
+-----------+
| 9.7.2     |
+-----------+
```

Any **9.7.x** is right. **8.4.x** is fine. **26.x** (an innovation release), **8.0.x**, or a version containing **MariaDB** means you are talking to a different server than you meant to: see the problems table below. Never "fix" a login with

`mysql_native_password`: that old method no longer exists in MySQL 9 (error 1524).

4 The course folder, and where the labs go

~2 min

Everything you write in the course lives in one folder, `mysql-course`, in your home folder. Lesson 1.3's `hello.sql` and Lesson 2.1's `create_bookstore.sql` are saved there, and so is the `labs/` folder your instructor hands out.

	WINDOWS (POWERSHELL)	MACOS / LINUX
Create it	`mkdir $HOME\mysql-course`	`mkdir -p ~/mysql-course`
Go there	`cd $HOME\mysql-course`	`cd ~/mysql-course`

```
mysql-course/
├── labs/                      handed out by your instructor: keep the whole folde
│   └── module-01/l1_0-orientation/check_environment.sql, .sh, .ps1
├── hello.sql                  (Lesson 1.3)
└── create_bookstore.sql       (Lesson 2.1) … and the other scripts you write
```

Put the three check files you received into `labs/module-01/l1_0-orientation/` inside `mysql-course` (create the folders; when the whole `labs/` folder arrives, it replaces them). **Always start the `mysql` client from inside `mysql-course`**: the lab command `SOURCE labs/…` looks for files relative to the folder the client was started in.

Do **not** load the course database `bookstore.sql` yet: you build `bookstore` yourself in Lesson 2.1, and Session 5 tells you when to download the full version. (Joining for Part 2 only? Then download `bookstore.sql` from the course home page into `mysql-course` now; your instructor sends Session 8's check instead.)

5 Run the environment check

~2 min

From a terminal in `mysql-course`, run the check for your system and type the root password when asked:

SYSTEM	COMMAND
macOS, Linux, WSL	`bash labs/module-01/l1_0-orientation/check_environment.sh`
Windows	`powershell -ExecutionPolicy Bypass -File labs\module-01\l1_0-orientation\check_environment.ps1`
Docker, client on the laptop	add the connection: `bash labs/module-01/l1_0-orientation/check_environment.sh -h 127.0.0.1 -P 3306 -u root -p` (the `.ps1` takes the same arguments)
Docker, no client on the laptop	`docker exec -i mysql97 sh -c 'exec mysql -uroot -p"$MYSQL_ROOT_PASSWORD" --table' < labs/module-01/l1_0-orientation/check_environment.sql` (PowerShell: `Get-Content labs\module-01\l1_0-orientation\check_environment.sql \| docker exec -i mysql97 sh -c 'exec mysql -uroot -p"$MYSQL_ROOT_PASSWORD" --table'`)

It checks that the `mysql` client is on your PATH and which version it is, then asks the server for its version, character set, SQL mode and login method, creates and drops a tiny test database, and looks for `bookstore`. The end of a good run on MySQL 9.7 looks like this:

```
+--------------------------------------------------+
| server_check                                     |
+--------------------------------------------------+
| OK     MySQL 9.7 LTS, the version the course uses |
+--------------------------------------------------+
…
+----------------------------------------------------------+
| bookstore_check                                          |
+----------------------------------------------------------+
| INFO   no bookstore database yet: Lesson 2.1 creates it |
+----------------------------------------------------------+
+-----------------------------+
| verdict                     |
+-----------------------------+
| READY for MySQL Foundations |
+-----------------------------+
```

✅ YOU'RE READY

Last table says **READY for MySQL Foundations**? Reply to your instructor with that table, and you are set for Session 1. **READY WITH NOTES** (MySQL 8.4 or MariaDB) is fine to start; the notes say what will differ. **NOT READY**? The line above it says what failed: try the table below, run the check again, then send the whole output.

COMMON INSTALL PROBLEMS

YOU SEE…	TRY THIS
Windows: `mysql` "is not recognized as the name of a cmdlet"	The MSI does not add the client to the PATH: Step 2, Windows point 4, then open a **new** PowerShell window.
macOS: `zsh: command not found: mysql`	Versioned Homebrew formulae are keg-only: the `export PATH=… mysql@9.7/bin…` line in `~/.zshrc`, then a new Terminal window (DMG: the `/usr/local/mysql/bin` line).
`SELECT VERSION;` says **26.x**	`brew install mysql` or `mysql:latest` gave you an innovation release. It works; for matching outputs reinstall `mysql@9.7` / `mysql:9.7`.
Ubuntu: version **8.0.x**, or root works only with `sudo mysql`	The distribution's own package. Add MySQL's APT repository (Step 2), or at least set a password with the `ALTER USER` statement in Step 3.
`ERROR 1045 (28000): Access denied`	The server answered and refused the name or password. Retype it (it is case-sensitive); "using password: NO" means you forgot `-p`. Docker: the password counts only on the first start of a volume.
`ERROR 2002` (socket) or `ERROR 2003` (port)	Nothing answered: the server is stopped, or you asked the wrong place. Start it (`Start-Service MySQL97`, `brew services start mysql@9.7`, `sudo systemctl start mysql`, `docker start mysql97`); with Docker use `-h 127.0.0.1 -P 3306`, not localhost.
`docker run`: "port is already allocated"	Another MySQL, MariaDB, XAMPP or MAMP listens on 3306. Stop it, or publish on another port: `-p 3307:3306`, then connect with `-P 3307`.
The check says **CHECK** … MariaDB	XAMPP or a distribution's default package. You can start with it; install MySQL 9.7 when you can.
Windows: the `.ps1` "cannot be loaded because running scripts is disabled"	Run it exactly as in Step 5, with `powershell -ExecutionPolicy Bypass -File …` (nothing to change system-wide).

YOU SEE…	TRY THIS
`ERROR 1524`: Plugin 'mysql_native_password' is not loaded	Old advice. Leave out `IDENTIFIED WITH mysql_native_password`; current clients and phpMyAdmin 5.2 use the default method.
A locked work laptop: nothing can be installed	Docker Desktop also needs administrator rights. Ask IT to install MySQL 9.7 before Session 1, or bring a personal laptop; tell your instructor.

Lesson 1.2's Common Mistakes section and the course's Troubleshooting page (troubleshooting.html on the course site) cover more cases, each with the exact error text.

THE COURSE AT A GLANCE

SESSIONS	YOU'LL LEARN
1–2	**Part 1.** What databases are, installing MySQL, the mysql client (and phpMyAdmin), creating databases and tables
3–4	Data types and constraints, adding and changing data safely, filtering, sorting and limiting (Checkpoint 1)
5–6	Aggregates and GROUP BY, subqueries, relationships and foreign keys
7	Joins, then the Part 1 assessment
8–9	**Part 2.** CTEs and window functions, indexes and views
10–11	Normalization, stored procedures and functions, transactions, JSON columns (Checkpoint 2)
12–14	Users and permissions, backups and imports, then your final project: a database you design, build, secure and back up

14 sessions of 3 hours in two parts (some programs run Part 1, Sessions 1 to 7, on its own). Every lesson uses the same fictional online bookstore: its books, customers and orders. Between sessions you finish each lesson's project, exercises, quiz and journal entry on the course site: about two to three hours a week.

FREQUENTLY ASKED

DO I NEED TO KNOW PROGRAMMING?

No. SQL is the language the course teaches, from the first statement. If you can use a spreadsheet and a file browser, you can start.

I ALREADY HAVE XAMPP, MAMP OR AN OLDER MYSQL. CAN I KEEP IT?

You can start with it: the environment check tells you what it is. XAMPP's database is MariaDB, and a few lessons (JSON operators, window and EXPLAIN details, some error numbers) will look different. For outputs that match

the pages, install MySQL 9.7, and stop the old server first so they do not fight over port 3306.

DO I NEED PHPMYADMIN OR MYSQL WORKBENCH?

No. The course uses the `mysql` command-line client, which comes with every install. Lesson 1.3 tours phpMyAdmin; Lesson 1.2 shows how to run it in Docker next to MySQL if you want it.

IS THE COURSE DATA REAL?

No. The bookstore's customers, orders and staff are invented for the course (the books are real classics). You build the database yourself, step by step, starting in Session 2.

WHAT IF A STEP FAILS AND I CANNOT FIX IT?

Come anyway, with the output of the environment check. Session 1 opens with the check and has an install clinic after the break, and you can pair with a neighbor until your machine is ready.

FOR THE FACILITATOR: ROOM SETUP

This section is for the instructor; it can stay on the learners' copy or be left off when you print it.

- ☐ **A week before Session 1:** send this guide with the three files from `labs/module-01/l1_0-orientation/` (`check_environment.sql`, `.sh`, `.ps1`); ask each learner to reply with the last table it prints. Keep a list: no reply, or NOT READY, means an install risk to chase before the day.
- ☐ **USB sticks or a shared folder** (the fallback when venue Wi-Fi fails): the `labs/` folder **without any** `*_solution.sql` files and without `checkpoint-*` and `part1-assessment` until they are used; and the **installers**: the Windows MSI (x86, 64-bit), both macOS DMGs (ARM and x86), the `mysql-apt-config` package, and for Docker users the image saved with `docker save mysql:9.7 -o mysql97.tar` (they run `docker load -i mysql97.tar`). A few hundred megabytes per learner over venue Wi-Fi is how an install clinic stalls.
- ☐ **Your own machine:** MySQL 9.7 with the same `mysql-course` folder; the environment check ends with READY (so you can show what it looks like). From the kit folder, `python3 labs/check_labs.py module-01` prints PASS on every line (point it at your server with the `MYSQL_LABS_*` variables in the README); run each session's check before that session.
- ☐ **Projector:** a terminal in `mysql-course` at 18 pt or more, logged out (you log in during the Lesson 1.2 demo), a second terminal already in the `mysql>` client, the session's deck from `slides/` and the lesson pages open in a browser.
- ☐ **The room:** a power socket for every laptop (bring power strips), a whiteboard, sticky notes (help flags and exit tickets), printed workbooks or a shared PDF, and Wi-Fi for the lesson pages. After Session 1 no session needs a download.
- ☐ **Install trouble on the day:** do not debug one machine in front of the room. Two minutes per machine during the opening check, then pair the learner with a ready neighbor; the install clinic after the break is for the rest.

PATH and a stopped server are one-minute fixes; a full download is not done on venue Wi-Fi. The Session 1 facilitator guide has the full trap table.

- ☐ **Locked work laptops:** every install path needs administrator rights (Docker Desktop too). Ask the learner's IT team to install MySQL 9.7 before Session 2, or pair the learner for the day; MySQL's Windows ZIP archive runs without installing but needs manual setup, so offer it only with your help.

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